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AADHAAR, PAN & PASSPORT PROCESS HELPER: An AI-Assisted Citizen Guidance System Using Retrieval-Augmented Generation (RAG)

Project ID:PRJO61

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SECTION 1: INTRODUCTION

1.1 Introduction

Government ID documents are of course of the utmost importance in the everyday lives of Indian citizens. The Aadhaar card, the PAN card and the passport are all used all the time for identity verification, for financial transactions, for accessing government services, for education, for employment and for international travel. Although the application processes for these documents are available via the official government websites, it's quite possible that users will find the whole process a bit bewildering. They may not be entirely clear on what documents are required, what application types are available, how to make corrections, what the eligibility criteria are, what the fees are and how long the processing will take. The Aadhaar, PAN & Passport Process Helper is a centralised digital assistance platform which provides information and initial guidance on the various identity document processes. The system uses AI and Retrieval Augmented Generation to answer your questions by pulling information from its knowledge base. In addition to the chatbot, it provides document checklists, name mismatch analysis, service information and user account functions.

1.2 Background of the Project

Traditional info searching involves going to a number of sites and reading long documents. Keyword based search systems can also fail to find the relevant information if the user's question is phrased in a different way to that which is used in the original document. Large Language Models can understand natural language questions and produce human readable answers. They can however be wrong in some instances as they don't automatically have access to the most up to date or specific documents that may be required for a given application. Retrieval augmented generation deals with that problem by retrieving relevant information

from an external knowledge base prior to generating a response. This project is the application of the RAG approach to identity document guidance. It's document retrieval coupled with Gemini for response generation to provide context-aware assistance.

1.3 Project Overview

The application focuses on three major categories of identity-document services:

Aadhaar Services

- Fresh Aadhaar enrolment guidance.
- Aadhaar demographic updates.
- Name and date-of-birth correction information.
- Biometric update information.
- Aadhaar PVC card guidance.

PAN Card Services

- Instant e-PAN guidance.
- New PAN application through Form 49A.
- PAN correction information.
- PAN-Aadhaar linking guidance.

Passport Services

- Fresh passport application guidance.
- Normal and Tatkaal scheme information.
- Passport reissue guidance.
- Non-ECR information.
- Police Clearance Certificate guidance.
- Required-document information.

The application also includes an AI chatbot, personalized checklist generation, name discrepancy analysis, user authentication and a dashboard for saved checklist progress.

1.4 Motivation

The main objective of this project is to minimize the confusion faced by citizens in the preparation of identity document applications.

Users may have different questions depending on their personal situation. For example, the documents required by an individual applying for a new passport will be different from the one required by a person applying for a passport reissue.

Differences in name spelling, initials or date of birth across identity documents may cause verification problems.

A centralized system can assist users to understand these requirements before visiting the relevant official portal or service centre.

1.5 Scope of the Project

The scope of this project includes:

1. Providing guidance for Aadhaar, PAN and Passport procedures.
2. Answering natural-language questions through an AI chatbot.
3. Retrieving relevant information from a knowledge base.
4. Generating document checklists based on selected application details.
5. Comparing names and dates of birth across selected documents.
6. Providing preliminary recommendations for possible discrepancies.
7. Supporting user registration and login.
8. Saving checklist progress in a user account.
9. Displaying relevant official government portal links.

The project is intended for information and guidance purposes. It does not directly submit applications, issue identity documents or retrieve live application status from government databases.

1.6 Organization of the Report

This documentation is divided into six major sections. 1. Background and Introduction to the Project Section 2 describes problem statement, existing system, proposed system and objectives. Section 3 describes the system architecture and the RAG workflow. In Section 4 the implementation and the technologies used are described. Section 5 presents the testing approach and the application results. Section 6 presents the conclusion and future improvements.

SECTION 2: PROBLEM STATEMENT AND OBJECTIVES

2.1 Problem Statement

Citizens applying for Aadhaar, PAN cards and passports often have to collate information from several government websites, circulars and application documents.

The data is often large, spread across multiple sources and hard to comprehend. Users may also not know what documents are needed for a particular service, or the procedure to follow when there are differences in personal details across identity documents.

Traditional keyword searches may not always understand the meaning of what a user asks. General-purpose AI chatbots can also give unsupported answers if they are not provided with relevant reference information.

Hence, there is a need for a centralized intelligent assistance system that can retrieve relevant identity-document guidelines, generate understandable responses, provide document checklists, and identify possible personal-detail discrepancies.

2.2 Existing System

The current approach typically requires users to sift through individual government websites, PDF instructions, FAQs and manual searches to learn how to apply.

A citizen, for example, may need to visit the UIDAI portal for Aadhar-related information, the Income Tax e-Filing portal for PAN-related information and the Passport Seva portal for passport-related information.

The user must select the appropriate procedure and identify the documents that apply to their particular situation.

2.3 Limitations of the Existing System

1. Information is distributed across multiple websites.
2. Users may need to read lengthy PDF documents.

3. Keyword searches may not understand the context of a question.
4. Users may find it difficult to identify the correct document checklist.
5. Name and date-of-birth differences may go unnoticed until the application process.
6. Users may need to repeat searches for similar questions.
7. General-purpose chatbots may provide information without showing the supporting source.

2.4 Proposed System

The proposed system is an AI based web application for providing centralized guidance for Aadhaar, PAN and Passport procedures.

It includes a RAG based chatbot, rule-based checklist generation, name discrepancy analysis and user account management.

The chatbot fetches relevant information from the knowledge base and feeds the fetched content to the Gemini model to generate a context-aware response.

The checklist module applies predefined conditions to identify documents relevant to the chosen service. The discrepancy analysis module compares the entered names and dates of birth and makes preliminary recommendations.

The app is designed to ease accessibility, minimize the need for manual search and assist users in preparing for official application procedures.

2.5 Objectives

The main objectives are:

1. To develop a centralized web application for Aadhaar, PAN and Passport guidance.
2. To implement RAG for document-based question answering.
3. To integrate a Large Language Model for natural-language response generation.
4. To use ChromaDB for storing and retrieving document embeddings.
5. To implement a keyword-based retrieval fallback mechanism.
6. To generate document checklists based on selected application requirements.
7. To identify possible name and date-of-birth discrepancies across identity records.
8. To provide recommendations for resolving detected inconsistencies.
9. To implement user registration, authentication and saved checklist functionality.
10. To provide a simple and accessible user interface.

2.6 Advantages of the Proposed System

- Centralized guidance for three important identity-document services.
- Natural-language question answering.
- Retrieval-based responses using stored reference information.
- Document checklist generation.
- Preliminary name discrepancy identification.
- User account and checklist tracking functionality.
- Access to relevant official government portal links.
- Fallback retrieval when the ChromaDB knowledge base is unavailable or empty.

SECTION 3: SYSTEM ARCHITECTURE – RAG PIPELINE

3.1 System Architecture Overview

It is designed as a modular system that incorporates a user interface, application logic, knowledge retrieval, response generation and database management.

The user interacts with the application via the Streamlit interface or the standalone HTML, CSS and JavaScript frontend. The application processes the user's request and invokes the appropriate service.

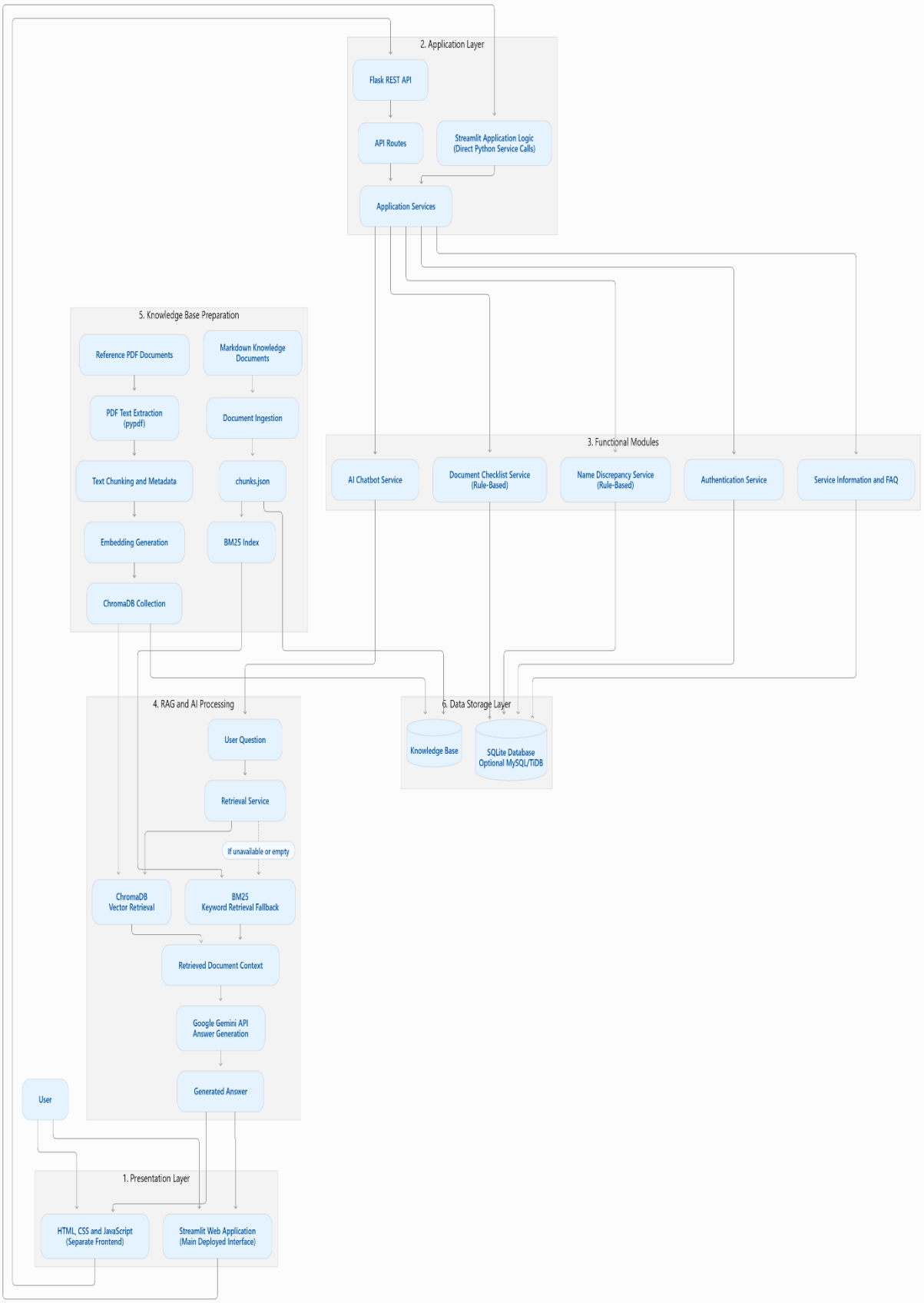
When responding to chatbot questions, the RAG service attempts to fetch relevant information from ChromaDB. If the ChromaDB collection is empty or unavailable, the application falls back to the BM25-based keyword retrieval mechanism.

The retrieved information is then passed to the Gemini API when the relevant generation path is available. The generated answer/fallback response gets displayed to the user.

Other functions, such as generating checklists and analyzing name discrepancies, are handled by their respective application services.

3.2 Architecture Flow

Overall System Architecture



3.3 Retrieval-Augmented Generation

Retrieval-Augmented Generation (RAG) is a method that combines information retrieval and response generation using Large Language Models.

Instead of relying only on what the language model already knows, RAG pulls in relevant information from an outside knowledge base and feeds that information to the model as context.

In this project, the RAG system is developed to answer the questions of Aadhaar, PAN and Passport procedures using the stored reference documents.

The RAG process consists of two major phases:

1. Data Ingestion Phase.
2. Query Processing and Generation Phase.

3.4 Data Ingestion Phase

The data ingestion phase prepares reference documents so that they can be searched efficiently.

The project contains two knowledge preparation approaches.

A. Markdown Knowledge Base

The existing knowledge-base module reads Markdown documents containing Aadhaar, PAN, Passport and name-mismatch guidelines.

The documents are divided into sections based on Markdown headings. The generated sections are stored as structured chunks in a JSON file named chunks.json.

These chunks are loaded into the custom Python retrieval component.

B. PDF-Based ChromaDB Ingestion

The newer RAG pipeline supports PDF-based ingestion.

The process includes:

1. Collecting PDF documents from the Aadhaar, PAN and Passport folders.
2. Extracting text from PDF pages using the pypdf library.
3. Dividing extracted text into smaller overlapping chunks.
4. Attaching metadata such as service category, document title, filename, official URL and page information.
5. Converting text chunks into numerical embeddings through ChromaDB's configured default embedding function.
6. Storing the embeddings and associated text in a persistent ChromaDB collection.

3.5 Query Processing Phase

When a user enters a question, the system processes it through the retrieval service.

In the ChromaDB retrieval path:

1. The user's question is received.
2. The question is converted into an embedding using the configured embedding function.
3. ChromaDB searches for similar document chunks.
4. The top four relevant chunks are retrieved by default.
5. An optional service filter can restrict retrieval to Aadhaar, PAN or Passport information.
6. The retrieved chunks are passed to the answer-generation module.

3.6 Large Language Model Integration

The project integrates the Google Gemini API for natural-language response generation.

The retrieved information and user's question are combined into a prompt and sent to the Gemini model.

The RAG generator instructs the model to:

- Answer using the retrieved evidence.
- Avoid inventing fees, timelines and eligibility conditions.
- Indicate when the available evidence is insufficient.
- Ignore instructions that may appear inside retrieved document content.
- Mention the supporting source identifiers.
- Direct users to official portals for confirmation.

3.7 Retrieval and Fallback Mechanism

The application contains two retrieval approaches.

ChromaDB Retrieval

ChromaDB is the vector database used by the newer PDF-based RAG pipeline. It supports embedding-based similarity search and retrieval of relevant text chunks.

BM25-Based Keyword Retrieval

The existing knowledge-base module uses a custom Python implementation of BM25 scoring. It tokenizes text, calculates term relevance and ranks document sections based on their matching content. It also provides an additional score when query terms appear in document titles.

The RAG service first attempts to use the ChromaDB pipeline. If the ChromaDB collection is empty or the pipeline is unavailable, it falls back to the BM25-based retrieval and response-synthesis service.

This fallback design allows the chatbot to continue providing responses when the newer vector database path is not available.

3.8 Document Checklist Generation Module

The checklist module generates a list of documents based on the selected identity-document service and application conditions.

The user selects the required service, such as fresh Aadhaar enrollment, instant e-PAN, physical PAN application or passport application.

The system applies predefined conditions and returns:

- Mandatory documents.
- Optional documents, where configured.
- Estimated fees.
- Estimated processing time.
- Relevant official portal.
- Additional notes and instructions.

The checklist is generated through rule-based Python logic. It does not depend on the Gemini model.

For passport applications, the logic can consider the selected Normal or Tatkaal scheme, availability of a 10th certificate and government employment information.

3.9 Name Discrepancy Analysis Module

The name discrepancy analysis module compares names and dates of birth entered for the 10th certificate, Aadhaar and PAN records.

The module first normalizes the entered names by removing selected titles, unnecessary special characters and extra spaces. It converts the names into a standardized uppercase format.

After normalization, the system performs several comparisons:

1. Single-name detection.
2. Initial-versus-expanded-name comparison.
3. Name-order comparison.
4. Partial name or spelling discrepancy detection.

5. Major name mismatch detection.
6. Date-of-birth comparison.

From the identified conditions, the module will identify possible problems, suggestions, and the severity level.

The severity levels include None, Low, Medium and High.

The module may also provide preliminary flags of required affidavits and required Gazette notices for certain conditions of mismatches.

This tool provides a preliminary analysis based on some set rules. The output does not necessarily mean a legal opinion.

3.10 User Authentication and Dashboard Module

The application provides registration and login functionality.

During registration, the system collects the user's full name, email address and password. The password is stored as a password hash rather than plain text.

During login, the system checks the supplied credentials and creates a session token.

The dashboard displays the authenticated user's details and saved checklist progress.

Users can save checklist progress, view the number of completed items and track the completion percentage.

3.11 Database Architecture

The project supports SQLite for local development and includes an optional MySQL/TiDB-compatible database connection.

The database contains the following main tables:

Services: Stores service codes, categories, descriptions, application modes, fee estimates, timelines and official portal links.

Acceptable Documents: Stores document names, proof types, categories, issuing authorities and notes.

Discrepancy Rules: Stores structured discrepancy-related information and recommendation fields.

FAQs: Stores frequently asked questions and their answers.

Users: Stores user details, email addresses, password hashes and roles.

User Sessions: Stores authentication session tokens and expiration information.

User Saved Checklists: Stores checklist progress associated with individual users.

SECTION 4: IMPLEMENTATION DETAILS AND TECHNOLOGY STACK

4.1 Programming Language – Python

Python is used as the primary programming language for backend development, RAG processing, checklist generation, name analysis, database operations and application logic. Its extensive libraries and support for artificial intelligence make it suitable for developing this application.

4.2 Streamlit

Streamlit is used to develop the main interactive web application interface. It provides the application pages and interactive controls for navigation, user login, chatbot interaction, Aadhaar guidance, PAN guidance, Passport guidance, FAQs and the dashboard. The Streamlit application is the main interface used by the deployed project.

4.3 Flask

Flask is used to implement the REST API backend. It provides endpoints for chatbot requests, service information, document checklist generation, FAQs, name discrepancy analysis and user authentication. Flask blueprints are used to organize different API functionalities into separate modules.

4.4 HTML, CSS and JavaScript

The project also contains a separate web frontend developed using HTML, CSS and JavaScript. HTML defines the structure of the pages, CSS controls the visual appearance and responsive layout, and JavaScript handles client-side interactions and communication with the Flask API. This frontend is separate from the Streamlit interface.

4.5 Google Gemini API

Google Gemini is integrated as the Large Language Model for generating natural-language responses. The Gemini API receives the user's question together with the retrieved context and generates a response. The newer RAG generator uses a low-temperature configuration to encourage more controlled factual responses.

4.6 ChromaDB

ChromaDB is used as the persistent vector database in the newer RAG pipeline. It stores document chunks and their embeddings, along with metadata such as service category, document title, filename, page information and official source URL. It supports similarity-based retrieval of information relevant to a user's question.

4.7 pypdf

The pypdf library is used to read PDF files and extract textual information from their pages during the PDF ingestion process. The extracted text is divided into smaller chunks before

being stored in the vector database. Scanned image-only PDF pages may require OCR support, which is not implemented in the current ingestion pipeline.

4.8 BM25 Retrieval

The project contains a lightweight custom Python implementation of BM25-based document retrieval. It calculates the relevance of document sections using query-term frequency, inverse document frequency and document-length normalization. The system also gives an additional relevance score to matching terms found in document titles. This mechanism is used as a fallback when ChromaDB retrieval is unavailable or the ChromaDB collection is empty.

4.9 SQLite and MySQL/TiDB

SQLite is used as the local database for storing application information, user details, sessions and saved checklists. The database layer also contains support for MySQL/TiDB-compatible cloud database connections through PyMySQL.

4.10 pytest

pytest is used for automated testing of the application's modules. The project contains tests covering authentication, chatbot functionality, checklist generation, service APIs, PDF ingestion, retrieval and answer generation.

4.11 Technology Stack Summary

Component	**Technology**	**Purpose**
Programming Language	Python	Core application logic
Main UI	Streamlit	Interactive web application
API Backend	Flask	REST API services
Additional Frontend	HTML, CSS, JavaScript	Web interface
Large Language Model	Google Gemini	Natural-language +generation
Vector Database	ChromaDB	Embedding storage and retrieval
PDF Processing	pypdf	PDF text extraction
Fallback Retrieval	BM25	Keyword-based document retrieval
Database	SQLite	Local data storage
Cloud Database Support	MySQL/TiDB	Optional cloud storage
Testing	pytest	Automated testing

4.12 Major Application Modules

Module 1 – Home and Navigation

Provides the main interface and navigation to the available identity-document services.

Module 2 – Aadhaar Guidance

Displays information about Aadhaar enrolment, demographic updates, biometric updates and PVC card services.

Module 3 – PAN Guidance

Provides information about instant e-PAN, physical PAN applications, corrections and PAN-Aadhaar linking.

Module 4 – Passport Guidance

Provides information about fresh passport applications, Normal and Tatkaal schemes, reissue procedures and supporting documents.

Module 5 – AI Chatbot

Accepts natural-language questions and provides responses through the retrieval and generation service.

Module 6 – Checklist Generator

Uses predefined conditions to generate a document checklist for a selected service.

Module 7 – Name Discrepancy Analyzer

Compares names and dates of birth and generates preliminary mismatch information and recommendations.

Module 8 – Authentication and Dashboard

Supports registration, login, session management and saved checklist progress.

Module 9 – FAQ and Process Guide

Provides frequently asked questions and step-by-step guidance for common identity-document procedures.

4.13 API Implementation

The Flask backend contains REST API endpoints for communication between the frontend and application services.

Important API routes include:

API Route	Method	Purpose
/api/health	GET	Checks backend availability
/api/services	GET	Retrieves service information
/api/faqs	GET	Retrieves frequently asked questions
/api/chat	POST	Processes chatbot questions
/api/chat/suggestions	GET	Retrieves suggested questions
/api/checklist/generate	POST	Generates a document checklist
/api/checklist/documents	GET	Retrieves acceptable document information
/api/services/analyze-mismatch	POST	Analyzes possible name discrepancies
/api/auth/register	POST	Registers a new user
/api/auth/login	POST	Authenticates a user
/api/auth/me	GET	Retrieves authenticated user details
/api/auth/logout	POST	Ends a user session
/api/auth/checklists	GET/POST	Retrieves or saves checklist progress

4.14 Security and Privacy Considerations

The application uses password hashing for storing user passwords and session tokens for authentication.

API requests are validated for required fields and message length. The chatbot route also validates the optional service filter.

Sensitive information such as API keys and database credentials should be stored in environment variables and should not be included in public source-code repositories.The project is an educational guidance application and should not be treated as an official government portal. Production deployment requires additional security configuration, secret management and validation.

SECTION 5: RESULTS AND TESTING

5.1 Introduction

Testing is an important stage of application development. It helps verify whether the individual modules and integrated system perform their intended functions. The project contains automated test modules for authentication, chatbot processing, checklist generation, service APIs, RAG ingestion, retrieval and response generation. The testing approach includes unit testing, API testing, module testing and integration testing.

5.2 Functional Testing

Functional testing is performed to verify the major features of the application.

The main functions tested include:

1. User registration and login.
2. Service information retrieval.
3. Chatbot question processing.
4. Knowledge-base retrieval.
5. Checklist generation.
6. Name discrepancy analysis.
7. FAQ retrieval.
8. Saving and retrieving checklist progress.

5.3 RAG Pipeline Testing

The RAG pipeline tests cover document preparation, chunking, retrieval and response generation. The ingestion tests verify text chunk creation, empty-page handling, deterministic chunk identifiers and metadata storage. The retrieval tests verify the returned document structure, empty knowledge-base handling, service filtering and error handling. The generator tests cover response generation, citation creation and fallback behaviour when the Gemini API is unavailable or returns an error.

5.4 Test Cases

The following table can be used in the final testing report. Update the actual result and status columns after executing the tests.

Test ID	**Test Scenario**	**Expected Result**	**Actual Result**
TC01	User registration with valid details	Account is created	To be recorded

TC02	Login with valid credentials	User is authenticated	To be recorded
TC03	Login with incorrect credentials	Error message is displayed	To be recorded
TC04	Ask an Aadhaar-related chatbot question	Relevant guidance is returned	To be recorded
TC05	Ask a PAN-related chatbot question	Relevant guidance is returned	To be recorded
TC06	Ask a Passport-related chatbot question	Relevant guidance is returned	To be recorded
TC07	Generate a passport checklist	Required document list is returned	To be recorded
TC08	Generate an instant e-PAN checklist	Relevant e-PAN requirements are returned	To be recorded
TC09	Compare identical names	No name discrepancy is detected	To be recorded
TC10	Compare different names	Possible discrepancy is identified	To be recorded
TC11	Compare different dates of birth	Date-of-birth issue is reported	To be recorded
TC12	Retrieve FAQs by category	Relevant FAQs are returned	To be recorded
TC13	Check backend health endpoint	Healthy status is returned	To be recorded
TC14	Retrieve information from an empty ChromaDB collection	Fallback mechanism is used	To be recorded
TC15	Save checklist progress	Checklist progress is stored	To be recorded

5.5 Sample Application Scenarios

Scenario 1 – Aadhaar Guidance
A user enters a question about correcting their Aadhaar name. The chatbot processes the question, retrieves relevant information from the available knowledge base and generates guidance.

Scenario 2 – PAN Guidance

A user asks how to apply for an instant e-PAN. The system provides relevant information about the application process and the documents or conditions represented in its knowledge base.

Scenario 3 – Passport Checklist

A user selects the Passport service and the Normal or Tatkaal scheme. The checklist module applies its predefined conditions and displays the corresponding document requirements.

Scenario 4 – Name Discrepancy Analysis

A user enters their name from the 10th certificate, Aadhaar and PAN. The system normalizes the values, compares them and displays possible differences and recommendations.

Scenario 5 – User Dashboard

A registered user logs in and views the saved checklist progress associated with their account.

5.6 Expected Results

The expected outcome of the project is a centralized application capable of providing identity-document guidance through an interactive web interface.

The RAG chatbot is designed to retrieve relevant reference information and generate understandable responses. The checklist module provides structured document requirements, while the name discrepancy module identifies possible differences in personal details.

The authentication module supports individual user accounts and checklist progress tracking.

The actual accuracy, response time and reliability of the system should be evaluated using executed tests and recorded observations rather than assumed values.

5.7 Limitations

1. The system depends on the quality and completeness of the stored knowledge base.
2. Outdated documents may lead to outdated guidance.
3. The checklist module uses predefined rules and stored fee and timeline estimates.
4. The name discrepancy module uses rule-based comparisons rather than a trained machine-learning model.
5. The current PDF ingestion pipeline does not include OCR for scanned image-only PDFs.

6. The application does not directly connect to government databases.
7. The application does not submit applications or provide live application-status tracking.
8. The availability of Gemini-based answers depends on API connectivity and valid configuration.
9. The actual chatbot retrieval mode depends on whether the ChromaDB collection is populated and available.

5.8 Screenshots to Include



Fig 5.8.1 – Home Page

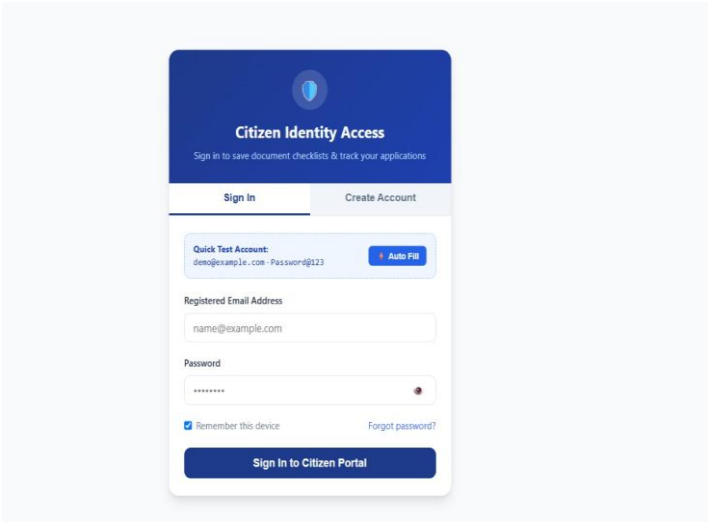


Fig5.8.2 – User Registration and Login Interface.

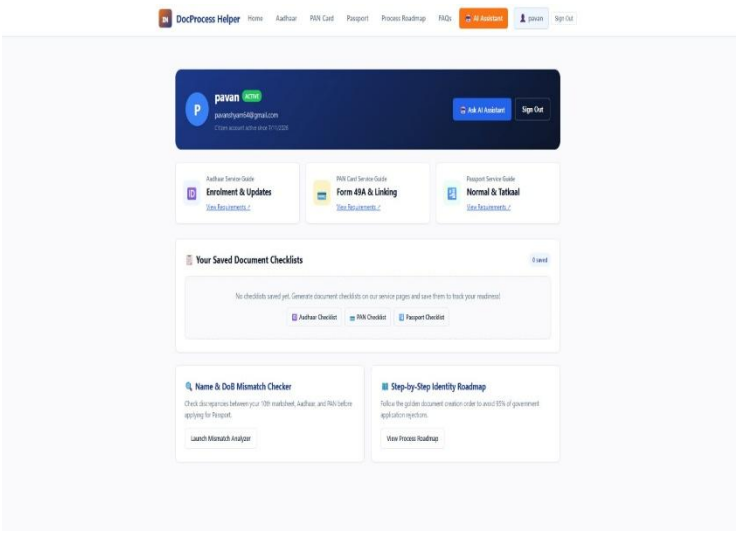


Fig 5.8.3 – User Dashboard.

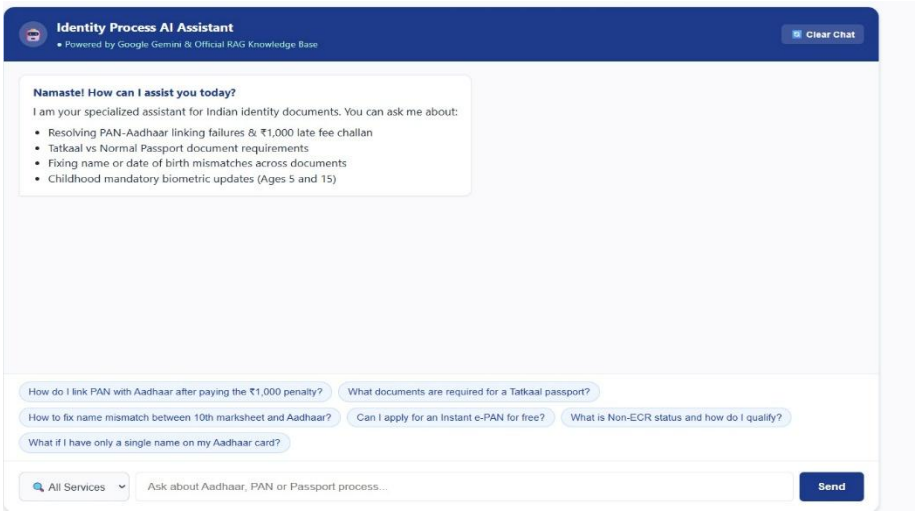


Fig 5.8.4 – AI Chatbot Interface.

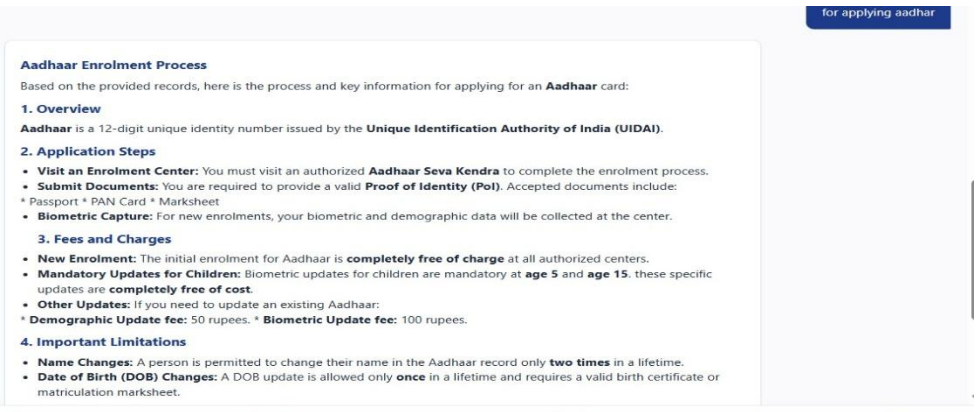


Fig 5.8.5 – Sample Aadhaar Guidance Response.

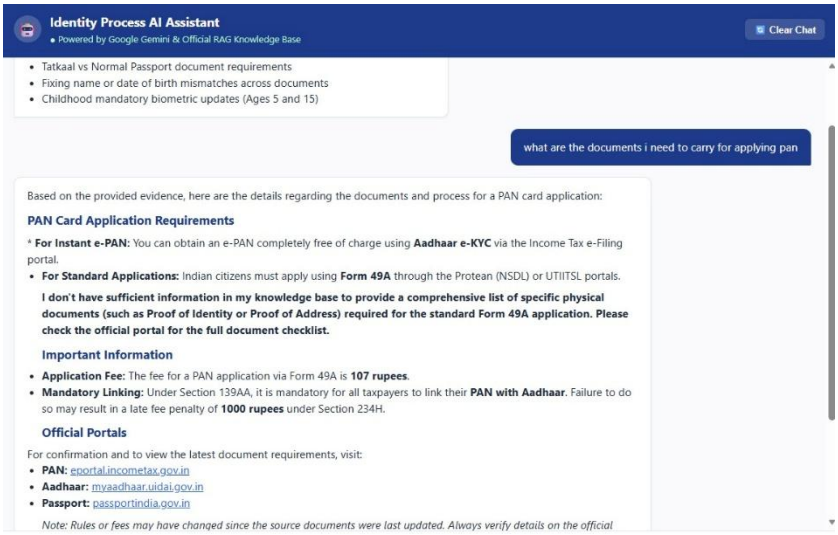


Fig 5.8.6 – PAN Card Guidance Page.

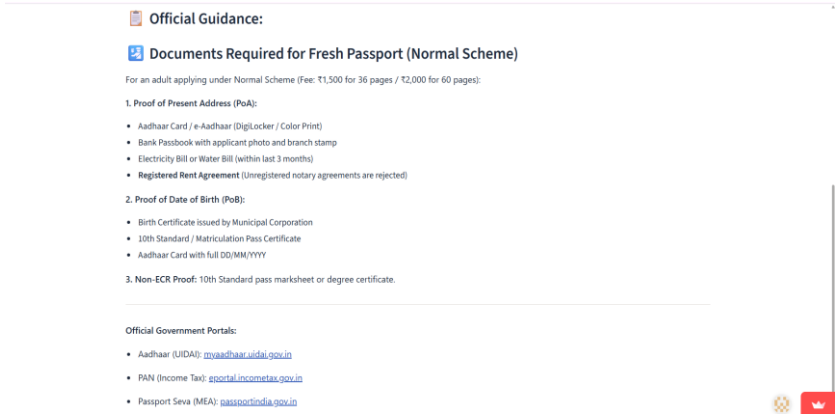


Figure 5.7 – Passport Guidance Page.

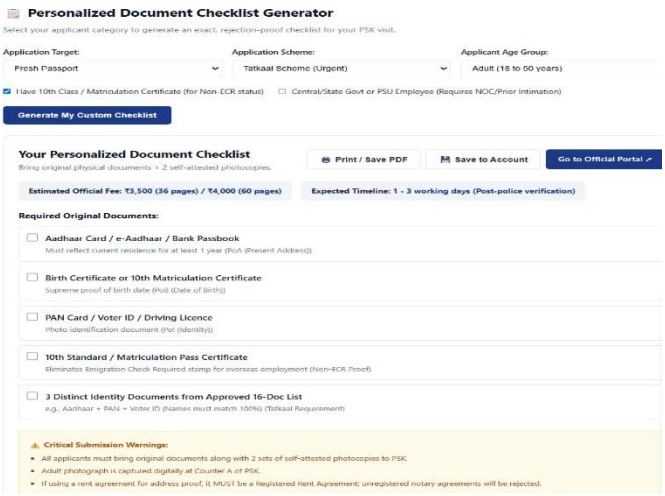


Fig 5.8.8 – Document Checklist Generation.

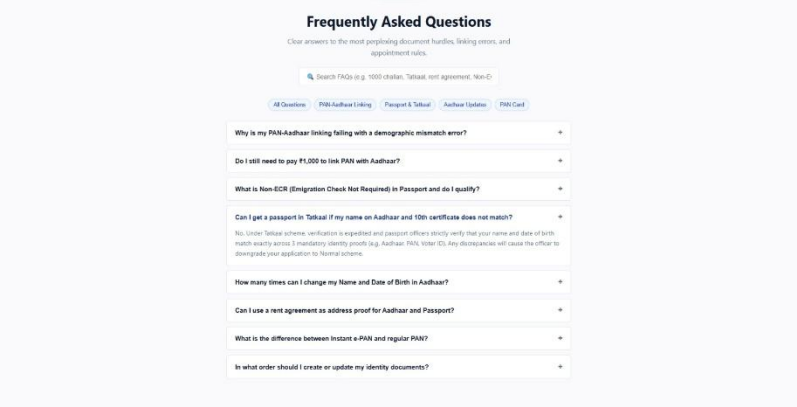


Fig 5.8.9 – FAQ and Process Guide Interface.

SECTION 6: CONCLUSION AND FUTURE SCOPE

6.1 Conclusion

Aadhaar, PAN & Passport Process Helper is an artificial intelligence aided guidance app created to make it easier for citizens to understand document application processes in India.

The application is a blend of Retrieval-Augmented Generation, large language model based response generation, rule based checklists, name discrepancy analysis and database management through a central web application.

The RAG based chatbot is created in such a way that it fetches relevant information from the knowledge base and generates context-based responses using Gemini API. ChromaDB facilitates the vector based retrieval in the new pipeline whereas BM25-based approach serves as a fallback retrieval approach.

Checklist module helps users find out the necessary documents for selected service. Name Discrepancy Analysis Module compares entered personal details and gives preliminary suggestions. Authentication of users and saved checklist module make it personalized.

In summary, the project demonstrates how artificial intelligence, information retrieval and rule based processing can be blended to create a useful citizen guidance application.

The application is meant to help users understand application processes and prepare their documents. However, final requirements and decisions have to be confirmed from the respective official government portal.

6.2 Future Scope

The project can be improved through the following enhancements:

1. OCR-Based Document Processing

Integrate Optical Character Recognition to extract text from scanned PDFs and image-based documents.

2. Real-Time Government Information Updates

Develop a mechanism to regularly verify and update stored guidelines, fees and application procedures using trusted official sources.

3. Multilingual Support

Add support for regional languages such as Telugu, Tamil and Hindi to make the application more accessible to citizens.

4. Advanced Hybrid Retrieval

Combine semantic vector search with BM25 keyword retrieval to improve the relevance of retrieved information.

5. Document Upload and Analysis

Allow users to upload supported identity documents and extract relevant details for preliminary comparison, with appropriate privacy and security controls.

6. Improved Name Matching

Enhance the discrepancy analysis module using advanced string similarity techniques and more comprehensive rules for initials, spelling variations and name ordering.

7. User Feedback Mechanism

Allow users to provide feedback on chatbot responses and retrieved sources to help evaluate and improve system quality.

8. Role-Based Access

Introduce separate access levels for administrators, users and document maintainers.

9. Mobile-Friendly Application

Improve the responsive interface and provide a better experience on smartphones and tablets.

10. Improved Security

Implement stronger secret management, secure session handling, input validation and production deployment configurations.

6.3 Final Outcome

The developed application provides a unified platform for identity-document guidance by combining AI-based question answering with structured rule-based services. It demonstrates the practical use of RAG, LLM integration, vector databases, backend APIs and database management in solving a real-world information-access problem.

REFERENCES

1. Unique Identification Authority of India (UIDAI) – https://uidai.gov.in/
2. My Aadhaar Portal – https://myaadhaar.uidai.gov.in/

3. Income Tax Department of India – https://www.incometax.gov.in/
4. Income Tax e-Filing Portal – https://eportal.incometax.gov.in/
5. Passport Seva Portal, Ministry of External Affairs – https://www.passportindia.gov.in/
6. Python Documentation – https://docs.python.org/
7. Streamlit Documentation – https://docs.streamlit.io/
8. Flask Documentation – https://flask.palletsprojects.com/
9. ChromaDB Documentation – https://docs.trychroma.com/
10. Google Gemini API Documentation – https://ai.google.dev/gemini-api/docs
11. pypdf Documentation – https://pypdf.readthedocs.io/
12. pytest Documentation – https://docs.pytest.org/

APPENDIX

A. Project Folder Structure

The project is organized into the following major directories:

- backend/ – Flask backend, database operations, APIs and application services.
- backend/rag/ – PDF ingestion, ChromaDB retrieval, generation and RAG configuration.
- backend/knowledge_base/ – Markdown documents, chunk generation and BM25-based retrieval.
- backend/database/ – Database schema, connection management and seed data.
- backend/routes/ – API route definitions.
- backend/services/ – Authentication, checklist, LLM and RAG services.
- frontend/ – HTML, CSS, JavaScript pages and assets.
- data/raw/ – PDF reference documents organized by service.
- data/source_manifest.json – Metadata and official source URLs for reference PDFs.
- tests/ – Automated tests for the main application modules.
- streamlit_app.py – Main Streamlit application entry point.

B. Abbreviations

Abbreviation	Expansion
AI	Artificial Intelligence

LLM	Large Language Model
RAG	Retrieval-Augmented Generation
API	Application Programming Interface
DB	Database
PDF	Portable Document Format
OCR	Optical Character Recognition
UIDAI	Unique Identification Authority of India
PAN	Permanent Account Number
MEA	Ministry of External Affairs
BM25	Best Matching 25
UI	User Interface
FAQ	Frequently Asked Questions
PoI	Proof of Identity
PoA	Proof of Address
PoB	Proof of Birth